annalen der physik
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Real-Time and Broadband Terahertz Wave Scattering Manipulation via Polarization-Insensitive Conformal Graphene-Based Coding Metasurfaces

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Here, for the first time, the real-time and broadband manipulation of terahertz (THz) waves are acquired by introducing a multifunctional graphene-based coding metasurface (GBCM). The designed structure consists of subwavelength patterned graphene units whose operational statuses can be dynamically switched between two digital states of "0" and "1". By engineering the spatial distribution of chemical potentials across the GBCM, various scattering patterns having single, two, four, and numerous reflection beams are elaborately achieved just within one planar structure. To compute the far-field pattern of GBCM, an inverse discrete Fourier transform (IDFT) is established, providing a fast and efficient design method. The proposed GBCM provides a low reflection bellow −10 dB over a broad frequency band ranging from 1 THz to 1.9 THz. In addition, the metasurface retains its low reflection behavior in a wide range of incident wave angles for both TE and TM polarizations. According to conformal invariance of graphene sheets, the stealth property of GBCM is well preserved while wrapping around a curved object. The proposed technique of real-time scattering manipulation leads to multifunctional THz devices, opening new routes contributing to numerous applications such as imaging and stealth technology.

1. Introduction

Terahertz (THz), whose frequency ranges from 0.1 to 10 THz, has attracted much attention due to having numerous applications such as communication, biological detection, and imaging.[1–5] To take advantage of all the benefits of THz waves, substantial efforts have recently been made to provide modern and multifunctional THz devices. The rapid expansion of metamaterial (MTM) technology is poised to offer feasible high-technological THz platforms that have been impressive in recent years.[6]

MTMs are artificial subwavelength metallo-dielectric structures, exhibiting extraordinary magnetic and electric features not found in nature. In the past decades, different behaviors such as negative or zero refraction index,[7,8] perfect absorption,[9] wave focusing,[10,11] and cloaking[12] have been reported for MTMs from microwave to visible frequency regions. Nevertheless, strong dispersion of resonant responses, lossy characteristics, and manufacturing complexities associated with three-dimensional bulky structures enable metamaterials to deal with severe practical constraints.[13] As an alternative, metasurfaces —densely arranged arrays of engineered inclusions— overcome the challenges encountered in their three-dimensional counterparts.[14] Their interactions with electromagnetic (EM) waves can be strong enough to attain very appealing functionalities while reducing fabrication complexities. More flexible control of EM waves along with higher spatial resolutions and more degrees of freedom are granted through employing metasurfaces.

The required bulky and complex materials introduced by conventional approaches such as transformation optics have long restricted the arbitrary manipulation of EM waves.[15] Nowadays, the wave-matter interaction of a flat metasurface, creating periodic gradient phase shifts in the range of $[0-2\pi]$, can behave anomalous scattering, yielding a generalized Snell's law.[16] Contrary to continuous phase-modulated metasurfaces, a variety of parameters such as shape, geometrical parameters, and constitutive materials can be engineered in an abrupt manner to have more profound control of EM waves. Recently, Giovampaola and Engheta have fostered the concept of coding metamaterials whereby the far-field scattering patterns of metallic objects can be simply controlled through elaborately designing the coding pattern of digital particles.[17] Afterward, numerous studies introduced single- or multi-bit configurations of coding metasurfaces in the microwave and THz spectra;[18,19] however, most of them have focused on a certain function such that the manipulation of EM waves is fixed once the design is completed.[20–29] More recently, some efforts have been made to realize programmable coding metasurfaces whose functionalities can be dynamically tuned at microwave frequencies through active materials.[30–32] However, as the frequency increases toward the terahertz frequency region, the pin-diodes is not commercially available so

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DOI: 10.1002/andp.201700310

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